derivatives of natural plasmids and are "engineered" to have features useful for cloning DNA.

Examples of plasmid cloning vectors : pBR 322 (plasmid discovered by Bolivar and Rodriguez 322) and pUC 19 (plasmid from University of California). Herbert Boyer and Stanley Cohen in 1973 showed it was possible to transplant DNA segments from a frog into a strain of *Escherichia coli* using pSC101, a genetically modified plasmid, as the vector. The work laid the foundation for the birth of Genetech, the first company dedicated to commercialization of recombinant DNA.

Figure 12.24a and 12.24b shows genetic maps of plasmid cloning vectors PUC19 and PBR322 respectively.

Plasmid cloning vector PUC 19 has 2,686 –bp and has following features:

1. It has a high copy number; so many copies of a cloned piece of DNA can be generated readily.
2. It has amp R (ampicillin resistant) selective marker
3. It has a number of unique restriction sites clustered in one region, called a multiple cloning site (MCS) or polylinker

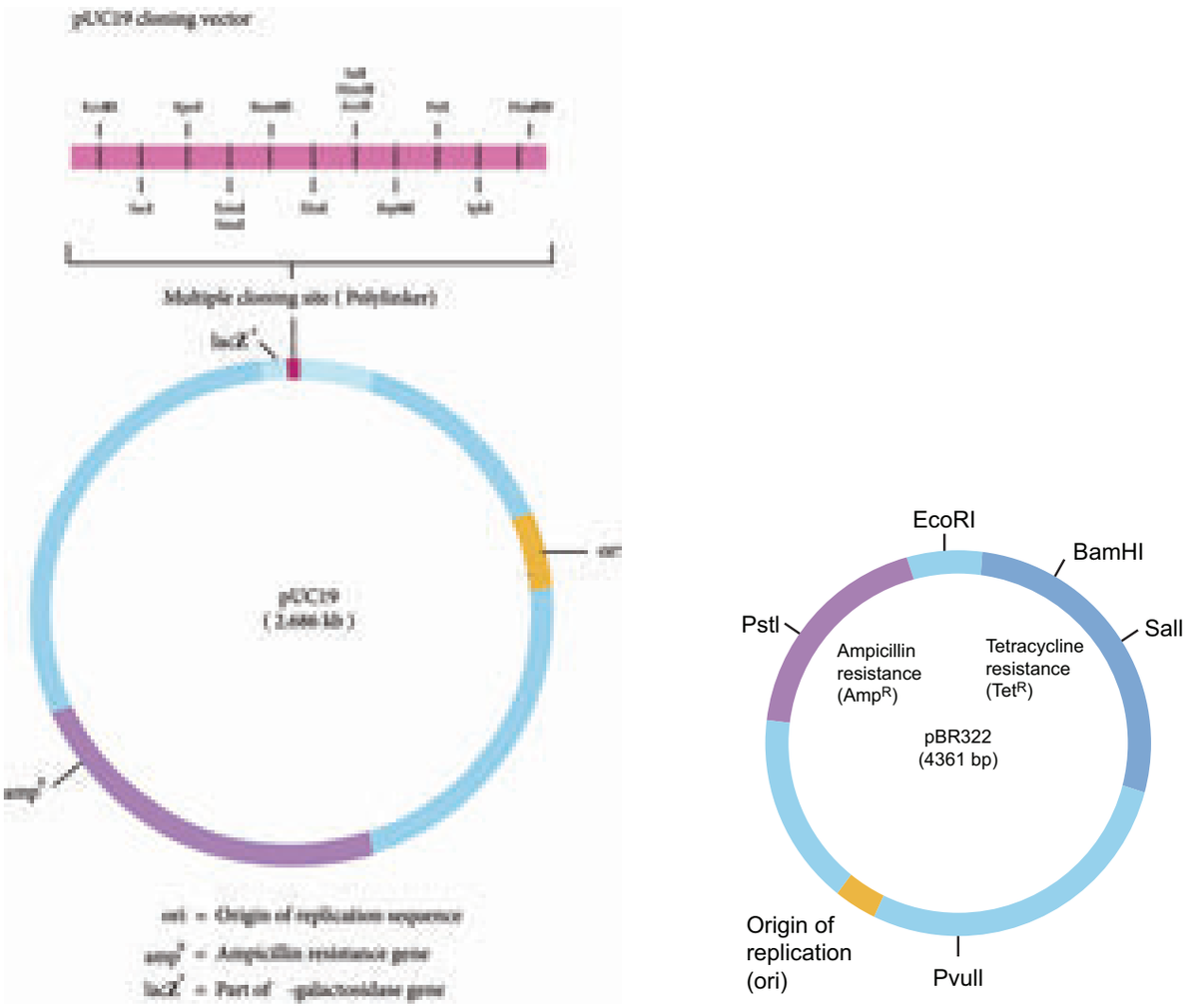


Figure 12.24: (a) PUC19 **(b)** PBR322

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